

INTERVIEW EXPERIENCES OF PGDBA BATCH-4

PRESENTED BY:

ALUMNI CELL, PGDBA BATCH-5

Anurag Malakar | Girish Chaurasiya | Jaiprita Prakash
Kunal Manchanda | Shashank Garg

D.E. Shaw

Selected 11 Candidates

9 Selected and 2 Rejected candidates shared their experience



Summary

◎ **Roles Offered:**

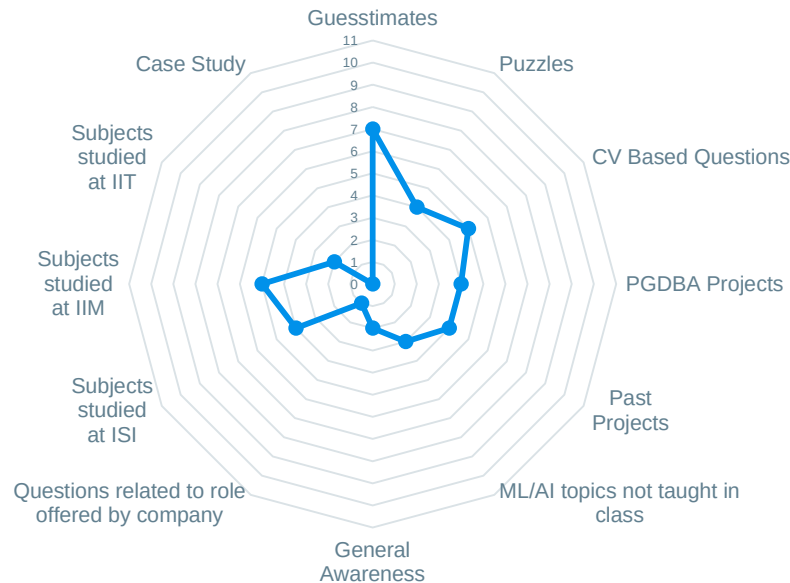
- Quant research intern (financial research)
- Marketing/post-trade analytics business intelligence intern (financial operations)

◎ **Shortlisting Criteria:** Resume Shortlisting

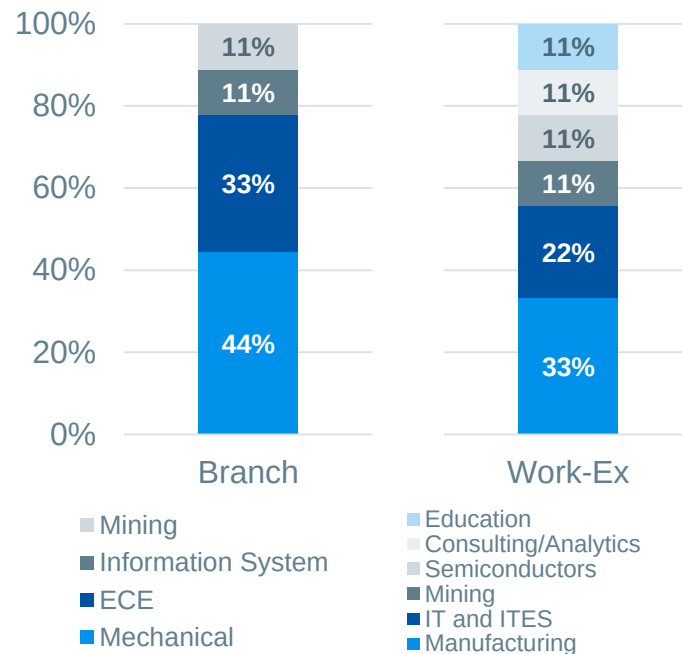
◎ **Average Interview Duration:** 30 minutes

◎ At least one technical round was taken

Topics asked in final interview



Profile of 9 Selected Students



Work-ex of 9 Selected Students



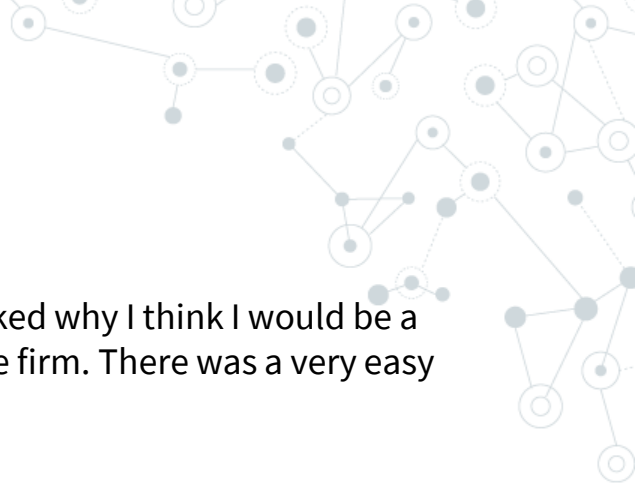
Saurabh Dubey

Tech Round 1 (Online Round)

Walk us through your resume and explain a project that you think we would be interested in knowing about, I chose my project on detecting causal impact using Bayesian structural time series. There were a couple of follow up questions on it like explain your forecasting methodology, how do you evaluate a time series model.

The next set of questions was based on my project on Time series forecasting using Hidden Markov models, they asked me to explain how HMM works and how do we interpret the latent states in an HMM. They also picked up a project that I had done on building a regression model for Final score forecasting in an ODI match. Questions asked were like how does your model take into account the recent change in the scoring trends in ODIs. They also asked me how I would model the given problem using Markov chains.

The final segment of the interview was on a puzzle where you had to find out the fastest 3 horses amongst 9 horses using a racing track with three lanes in the minimum number of races.



Saurabh Dubey

Final Interview (In person Round)

They explained the roles on offer and asked for my preference. They asked why I think I would be a good fit for the role and whether I had interacted with any seniors at the firm. There was a very easy puzzle on data imputation after that.

HR Question: What role would you prefer?

Additional Information

My interview luckily happened to be fairly technical but most other interviews had a very dominant case component where people were given a fictitious data set and had to come up with ways to generate revenue for DE Shaw. Resume based questions would mostly be on regression and time series projects. You should be well versed with your fundamentals and must be able to clearly explain the techniques used. Also, keep your CGPA as high as possible.



Vignesh

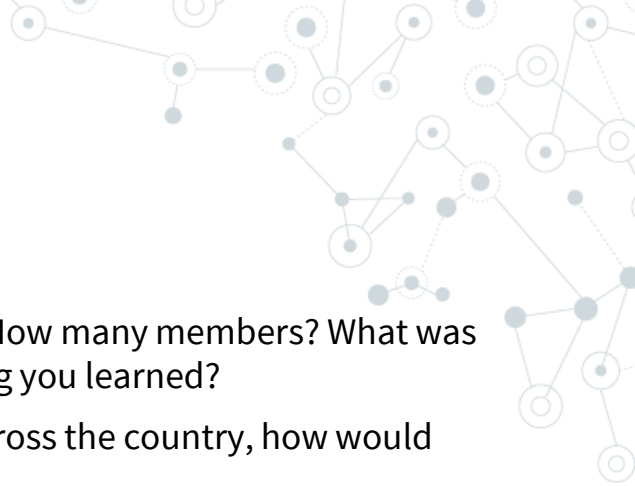
Tech Round 1 (Online Round)

- 1) Walk me through your resume
- 2) Describe the sentiment analysis project that you worked on
- 3) Could you speak about your work experience
- 4) If you are given access to the call and text data of all the mobile phone users, how will you be able to make money for us? (A lot of follow up questions)
- 5) 2 envelopes problem

Final Interview (In person Round)

You are expected to find target investors in Australia, Japan and China. Target investors could be Pension funds, Corporates or HNIs. Walk us through the approach that you would follow.

HR Question: None. However my interest in the role was gauged based on my reactions while they described the role.



Parkavi P

Tech Round 1 (Online Round)

- 1) Describe one project that you really liked. Was it a group project? How many members? What was your part? What were the challenges faced? What was the one thing you learned?
- 2) Given the video feeds from all parking lots of all shopping malls across the country, how would you make money for DE Shaw?
- 3) Why PGDBA given you're from the education field?
- 4) How would you select the 2/3 fastest horses from 9 with a lane size of 3?
- 5) Do you have any questions for us?



Continued on next slide...

Parkavi P

Final Interview (In person Round)

It started with any questions I had for them. I asked some questions about the roles offered and it went on for about 15 minutes.

Then I was asked this guesstimate question: How many people fly in and out of Kolkata in any time frame of your choice? They gave me a few minutes to think, helped me out a lot and nudged me in the right direction throughout. I think they were just checking whether I could pick up their clues. This went on for another 15 minutes and then the interview concluded.

HR Questions: None

Additional Information

They are just looking for a meaningful conversation. Practice cases, puzzles and guesstimates. While explaining projects, focus on the business aspects, dataset, and your learning and keep the technical stuff on the lower side unless you're asked specific questions on it. Try to follow their hints when you're missing something.



Anurag Saraf

Tech Round 1 (Online Round)

Regression and time series basics, imputation methods

Final Interview (In person Round)

They asked basic difference between regression models including and excluding constant terms. They asked where each of them is used and how to decide that. On data imputation, during PI round they provided a toy dataset and asked me to fill the missing data values. I suggested different techniques like KNN based and regression based imputation. They also gave another time series toy dataset and asked me to decompose that series that into it's trend and seasonal components. There was no error term involved and hence it was more or less straightforward.

HR Question : They asked me about my preference for the department at DE Shaw.

Additional Information

Institute rank and strong background in maths and statistics developed during ISI semester helped a lot in the interview.



Vanya Sinha

Tech Round 1 (Online Round)

- 1) Walk us through your CV and explain any project.
- 2) Tell an application of the project you just mentioned in finance.
- 3) Some work ex thing
- 4) You have audio data of calls between DE Shaw and investors/potential investors. How would you use this data? What will your approach be? Would you first convert the audio data to text and do analysis? Do you think the length of the call matters? What other variables can you look at?
- 5) Puzzle: 9 horses, 3 tracks. What's the min number of races you'd have to conduct to find out the second-fastest horse?
- 6) Any questions for us?

Vanya Sinha

Final Interview (In person Round)

Financial Research

- 1) Among the 7 offered, which DE Shaw roles would you prefer?
- 2) Marks of 6 students in 3 subjects is giving with a few values missing. At max, per student only one subject's marks are missing. How will you impute the values? Asked me at least 5 ways of doing this listing disadvantages of each method.
- 3) Outlier vs high leverage points. "Which one is more important?", "Which one would you remove?"

Financial Ops

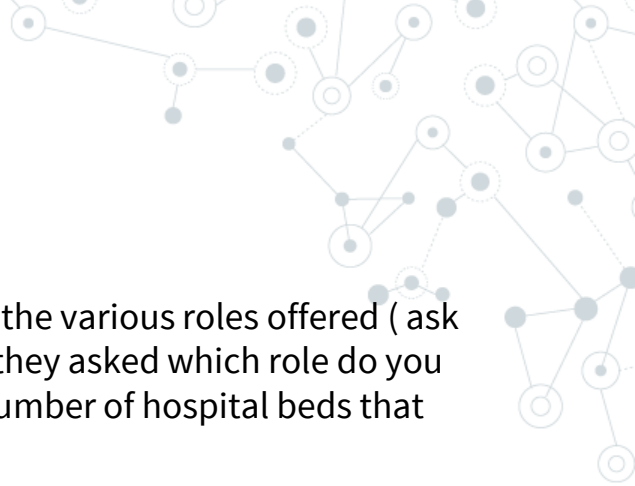
Given 3 regions - Australia, Japan, and China. You are to identify investors in these markets. How would you go about it? Consider types of investors, what information is publicly available etc. Very case-type interview.

HR Questions: None

Akash Vennamaneni

Tech Round 1 (Online Round)

- 1) Run through my career and what are my aspirations .
- 2) Followed by an investment question. It goes like this " You are given video traffic data , how do you come up with insights that will help de Shaw make investments " , there were many cross questions on my answers to this question.
- 3) After that I was asked two envelope puzzle (Just search it online you will be able to find the problem statement , You have many variations of the same puzzle and interviewer tried to confuse me with these variations)



Akash Vennamaneni

Final Interview (In person Round)

First question interviewer asked me is if I have any questions related to the various roles offered (ask as many questions as possible here) , once I stopped asking questions they asked which role do you prefer. After this they asked me a guesstimate. I had to estimate total number of hospital beds that are present in Kolkata and also estimate what should it actually be?

HR Questions: Regarding my choice of profile and why ?

Kolli Kiran Kumar

Tech Round 1 (Online Round)

- 1) Tell me about yourselves.
- 2) Elaborate a particular project mentioned above.
- 3) Revenue Generation Question : As I had work experience in Health Care Domain Company, I was asked "If I was given an hypothetical data set containing all billing information of hospitals, How can I create revenue for D.E. Shaw?"
- 4) Cross questions on revenue opportunities told. They will be checking if you have a solid reasoning for your answers.
- 5) Puzzle : Two Envelope problem
- 6) Any questions do you have for us?

Kolli Kiran Kumar

Final Interview (In person Round)

I had 2 rounds of interviews

1st Interview:

- 1) Any questions regarding the Roles offered. I got the roles clarified
- 2) STL Decomposition Question: Quarterly data was given over 4 years and asked to do the STL decomposition from the data. The interviewer wants to see if the trend & Seasonality were identified correctly.
- 3) Question on outliers. How do you identify an outlier? How does it affect the regression line?
- 4) Give an example when regression line without intercept is used. How does its R-Square differ when compared to normal regression line R-Square?
- 5) Puzzle: 9 Horse puzzle

2nd interview:

- 1) Guestimate Question : "Estimate the number of Spectacles sold in Kolkata in a day"
- 2) Any preference of roles offered.

HR Questions: None

Vidyadhar Mudium

Tech Round 1 (Online Round)

- 1) Questions based on work experience - asked to describe the project and then cross questions based on the answers.
- 2) Basic questions based on Linear Regression:
 - When do you use regression model with constant term and without constant term - give practical examples.
 - What is the significance of constant term.
 - Let's say you fit two regression lines with same data - one with constant and one without constant. Are the coefficients same in both the cases. If not, are there any cases when they will be the same.
- 3) One puzzle (with 5-6 seconds limit per answer):

You have 9 red socks and 11 blue socks in a dark room,

 - how many minimum number of socks you need to draw to guarantee getting a pair?
 - How many minimum number of socks you need to draw to guarantee getting a red pair?
 - How many minimum number of socks you need to draw to guarantee getting a blue pair?
 - How many minimum number of socks you need to draw to guarantee getting a red and blue sock?

Vidyadhar Mudium

Final Interview (In person Round)

First started with department I would be interested etc. Asked about whether I am aware of the roles offered. Then one case and then questions based on regression were asked.

- 1) Case: Post 2008-09 in USA, the oil consumption had started declining till 2014. What could be the reason. (Post Financial crisis it was a growth phase, everyone would expect that oil consumption in the country should go up, but it was going down.)
- 2) Regression based question: Questions based on difference between leverage point and outlier. Gave multiple cases to identify which point would have more impact on the regression line.
- 3) Since I was in Placecomm, I was asked about the experience, how it affected the preparation etc.

HR Questions: None

Additional Information

Regression and Time series, Puzzles mostly repeating from previous years.

Avi Sharma

Tech Round 1 (Online Round)

Q1: Related to a project mentioned in the CV : briefly explain the PS, goals of the project and methodology used.

Q2: **A scenario:** If given all the transaction data of various e-commerce websites in a given market, how will you use that data for DE Shaw as a firm which invests in various asset classes along with the hedging business.

Q3: **Another scenario:** What all parameters, attributes would you consider to understand/predict the demand of Aviation Fuel , given that you have to hedge on the pricing of Aviation fuel.

Q4: There are 25 horses. What is the minimum number of races needed so you can identify the fastest 3 horses? You can race up to 5 horses at a time.



Avi Sharma

Final Interview (In person Round)

Q1: Key challenges faced in the previous company.

Q2: Scenario Question : Given that you have all data related to clients of a competitor of a DE Shaw, how can you convert them as clients of DE Shaw, assume you have all the data.

Q3: Another Scenario Question : Given that you have a software that algorithmic ally does inter day trading and the law enforcing agency suspects of insider trading due to huge margins made, how will you prove that trades are not based on insider trading or any information which only your firm is privy to.

Q4: Question based on four different Time series data shown.

HR Questions: Which department are you interested in and would want to join if being selected?

Vikas Rai

Tech Round 1 (Online Round)

- 1) Hypothetical situation - You have GPS data of all cars. Generate Revenue out of it. Be creative and try to relate it with your work experience or any of your projects. Follow up questions on the same.
- 2) Puzzle :13 red socks and 17 blue socks. What is the minimum no. of withdrawal to guarantee a pair of socks. What is the minimum no. of withdrawal to guarantee a red pair of socks / blue pair of socks/mixed color pair of socks.

Vikas Rai

Final Interview (In person Round)

Finance Operations role

Tell me about yourself. Basic questions on CV. Which project do you like most? Describe it.

Situation: Algorithm does trading, so if one trade pair is such that it sells stock of a security like Reliance at say Rs. 100 in very large amount, this will create unrest in market and Reliance Bond prices will be lower, this algorithm will then buy these bonds to square off the position. This is called market manipulation and is not allowed. Questions are How will you detect this situation as there are million of trades its like needle in hay stock? What should you do to avoid this situation? Do you have any questions for us?

Finance Research Role

Do you have clarity of roles or would you like us to explain? Question: 5 students marks in subjects are given. 2 different students marks are not available in 2 different subjects. What is the best approximation? (Use KNN) Do you have any questions for us?

Additional Information

DE Shaw shortlists everyone. So prepare well. Going through last year internship experience and your projects, knowledge about finance trading are important. DE Shaw prefer CGPA, so have a good CGPA.



Shilpi Jaiswal

Tech Round 1

- 1) Tell me about yourself? What was your role in the previous company?
- 2) Discussion on 2 envelope puzzle
- 3) Any analytics projects you have worked on during PGDBA that is not taught in curriculum?
- 4) There are two states and different pipelines are going from one state to another. Water is flowing in different pipelines and there are some leakages in the pipe. How will you determine that in how many days water will reach from one state to another given that you know the time lags. Also suppose that there are 5 pipelines & 15 leakages in each pipe. What will be the complexity if u design an algorithm for this and how can you optimize the complexity?

MasterCard DS

Selected 12 Candidates

9 Selected candidates shared their experience

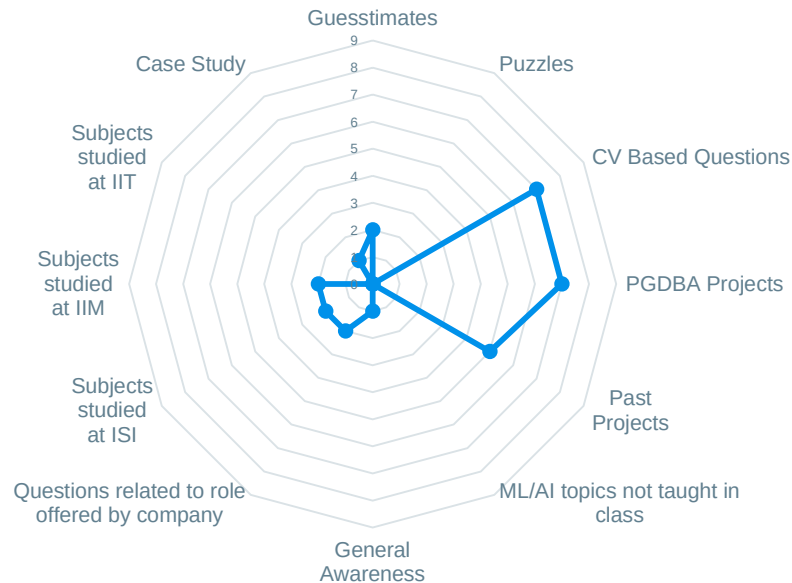




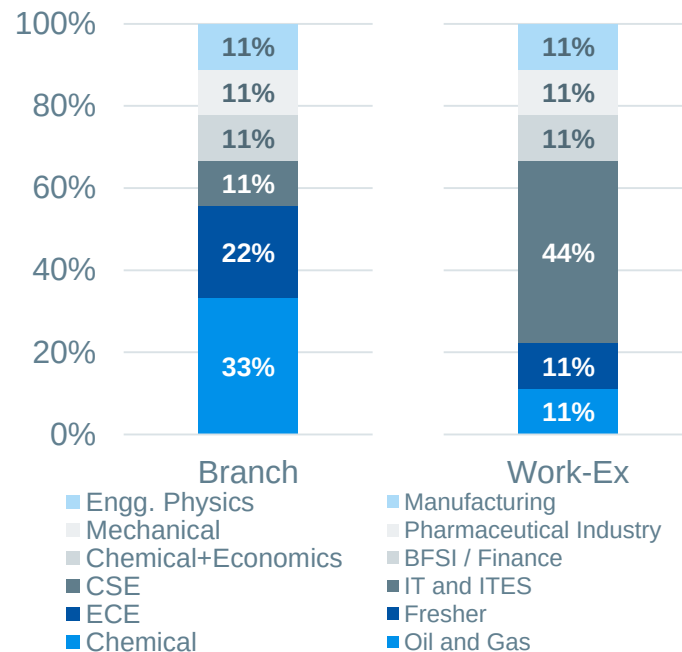
Summary

- ◎ **Shortlisting Criteria:** Resume Shortlisting
- ◎ **Average Interview Duration:** ~20 minutes
- ◎ Interview duration ranged from 5 minutes to 40 minutes.

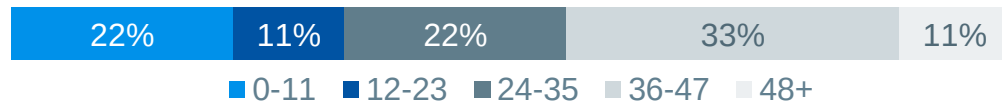
Topics asked in final interview



Profile of 9 Selected Students



Work-ex of 9 Selected Students



Abhishek Sharma

Technical Round

- 1) How do you apply PCA?
- 2) What are eigen vectors?
- 3) How do u determine K in K-means?
- 4) Explain hidden states in HMM??(My project in HMM)
- 5) What are latent factors in Collaborative Filtering

Final Round

They focused on the projects done and interview hovered around them.

Asked little bit on previous work experience as my one year was relevant. Being thorough with CV is really important.

HR Questions

Why PGDBA?, What does MasterCard do?, How did you add value in your last organization?



Prasanna Manickavasagan

Final Round

I was asked about a project on my resume

HR Questions

Location related questions

Additional Information

The interview was primarily a formality and they already had a list of candidates that they were keen to hire

Rishav Saha

Final Round

Majorly HR based questions. Hardly any technical question was asked.

HR Questions

- 1) Explain mastercard's business model
- 2) Difference between the business models of mastercard and AMEX
- 3) Why PGDBA after CSE?
- 4) Weakness and what are you doing to overcome it
- 5) Hobbies
- 6) Where do you see yourself in a few years
- 7) Challenges you faced while solving any of the problems in your projects
- 8) Business news headline in the last 10 days

Additional information

Focus on analytics/ business competitions & try and keep a CG above 9 (NOTHING ELSE MATTERS)

Vikas Rai

Technical Round

- 1) According to my resume I was told to describe my work project in objective problem format like What is your objective? What is hypothesis being tested? Your test statistic and all ? I described my fuel economy improvement question?
- 2) Then he asked after learning of PGDBA What better you could have done in your job profile? Described warranty tracking problem?
- 3) 2nd person asked me about ADAM (one of my project) very brief , XgBoost , Difference between LightGBM and XgBoost? He gave me a python Coding question to solve related to groupby. Then Asked me if I have to include more than 2 attribute in the same like mean and max ?

Vikas Rai

Final Round

- 1) Tell me about yourself?
- 2) Did you attend the Mastercard PPT? What all did you see projects?
- 3) What do you know about Mastercard?
- 4) Then he gave me a puzzle.
- 5) Four people come to a river in the night. There is a narrow bridge, but it can only hold two people at a time. They have one torch and, because it's night, the torch has to be used when crossing the bridge. Person A can cross the bridge in 1 minute, B in 2 minutes, C in 5 minutes, and D in 8 minutes. When two people cross the bridge together, they must move at the slower person's pace. What is the time taken?
- 6) Do you have any questions for us?

Vikas Rai

Additional Info

- 1) Mastercard shortlisted all people in 2 roles. There were no overlaps. Regarding projects . Diversify your projects as much as possible NLP, Recommendation System, Image, Classical Machine Learning. All these projects should solve a business problem.
- 2) Other Suggestions:
 - Maintain your GPA.
 - Prepare your work ex before hand. Include Data related terms more. If you have analytics experience, write well in your CV. Most companies are looking for prior analytics experience.
 - Describe and diversify your projects well
 - Focus on your coding skills, most companies have this as shortlisting criteria.

Karan Raj Chauhan

Final Round

- 1) Puzzle: How will you cut a cake to make 8 equal pieces with icing intact, cutting it only 3 times
- 2) Puzzle: Four people on a rickety bridge
- 3) General Awareness: What is wrong with the economy?,
- 4) Education: Why leave physics, Questions about minor degree

HR Question: Why do you want to join us

Rahul Mehta

Final Round

- 1) What does MasterCard do?
- 2) What did you do during your internship at Credit Suisse?
- 3) Where are you coming from? (I said PwC). Did you prefer PwC over us?

HR Questions

- 1) Two weaknesses that your friends will say that you have.
- 2) Did you join PGDBA out of boredom at your previous job?
- 3) What languages do you know about?

Shrustijit Sahoo

Technical Round

Questions were asked around Decision Trees, RandomForest, Gradient Boosting, Linear Regression, LightGBM, Cross Validation Techniques. I was also asked to explain link between PCA and SVD in simple terms.

Business round

A case was given to evaluate the parameters to be considered while opening a new shopping mall in a metropolitan area(eg. customer demographics, customer sentiment analysis etc).

HR Questions

- 1) Who is MasterCard's biggest competitor? (Ans: Cash and not Visa/Paypal/Rupay)
- 2) Have you ever been outside India?
- 3) How will you adjust in Delhi-NCR?

Rajath Nandan

Final Round

- 1) Case Study - There has been a decrease in revenue recently for an air service provider. How to identify the problem/caus given you have data of ticket sales data for the airline industry.
- 2) Picked time series project & asked me to explain in detail.

HR Questions

- 1) Previous work experience related questions & asked why I left my previous job.
- 2) What does MasterCard generate revenue or what are the services offered by MasterCard.
- 3) Related to relocation & adaptability etc.

Shilpi Jaiswal

Final Round

- 1) What analytics projects have you worked on during PGDBA?
- 2) What was your role in the previous company?
- 3) How is the work in your previous company different from the one you will going to do in Mastercard?
- 4) What does Mastercard do?
- 5) Are you comfortable with working on Tableau?
- 6) Have you worked on any report making tool?

HR Questions

- 1) Tell me about yourself?
- 2) Why did you join PGDBA?

MasterCard AI

Selected 7 Candidates

5 Selected candidates shared their experience

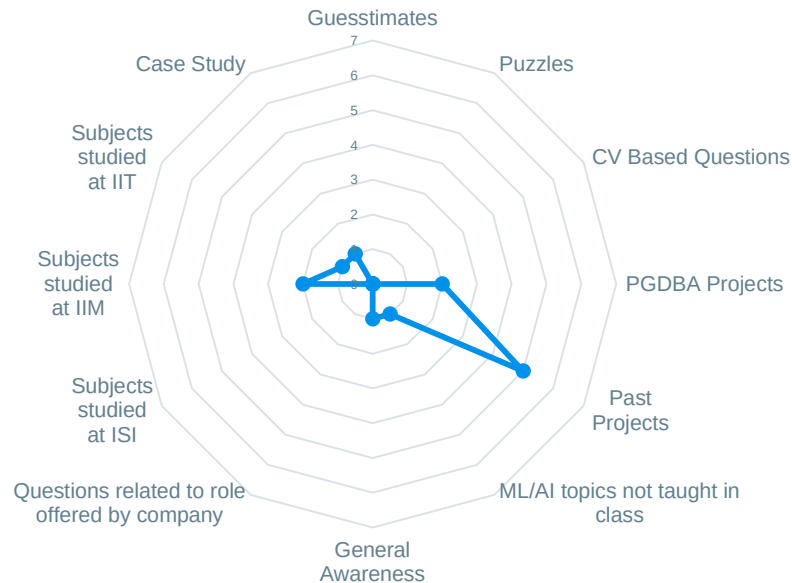




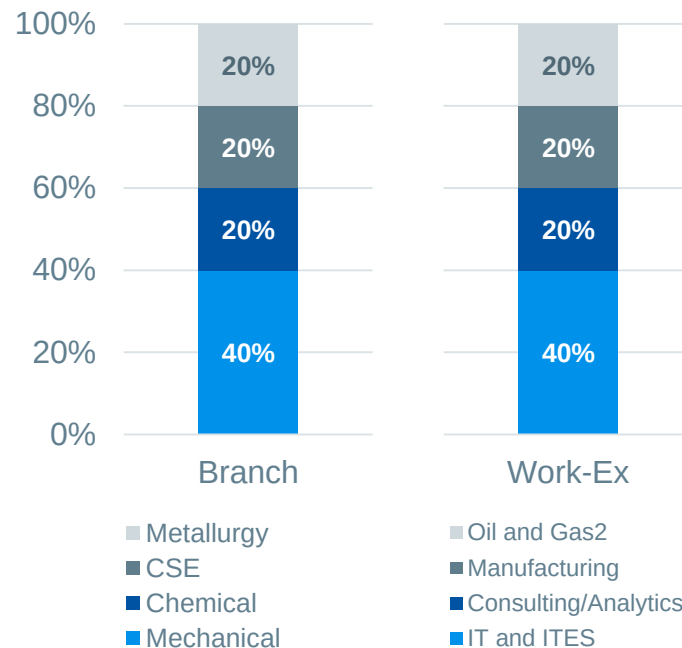
Summary

- ◎ **Shortlisting Criteria:** Resume Shortlisting
- ◎ **Average Interview Duration:** ~20 minutes
- ◎ Interview duration ranged from 5 minutes to 45 minutes.
- ◎ Out of 5 candidate, only 1 had to face technical round

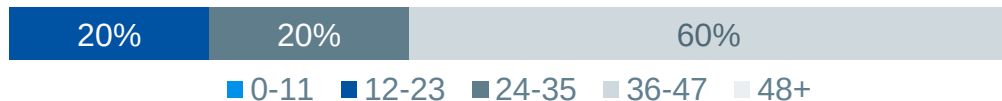
Topics asked in final interview



Profile of 5 Selected Students



Work-ex of 5 Selected Students



Aniruddha Mitra

Final Round

- 1) Explain IR project.
- 2) How random forest perform on text classification
- 3) Feature selection algorithm details
- 4) Recommendation algorithm: How to manage situation where the environment changes with time.
- 5) How to detect structural break in time Series. If the break occurs over a comparatively long period, how to manage.
- 6) Model checking in Time Series (likelihood vs others)
- 7) How to choose break point in ROC



Kanishka Kayathwal

Final Round

I was asked to explain any one of my PGDBA project

Nitish Kumar

Final Round

It was a very short interview and looked almost pre decided. They asked me to explain a project and I explained for almost 15 mins. In between they were asking very basic questions from the project as to how and why.



Garima Arora

Final Round

Questions asked originated from my resume plus a couple of general data pre-processing questions:
How is One-Hot Encoding done? Describe your favorite project.

HR Questions

How long did you work on your startup? Was it a solo stint?

Suhas Powar

Final Round

- 1) Asked if you have published any paper in your college time?
- 2) I did mentioned about my thesis in M.Tech
- 3) Explain ELMO, Bert, Word2vec. What are the difference between them?
- 4) What all variable you will use for weather forecasting which model you will use?
- 5) Explain Newton-Raphson method of optimization (it was mentioned in my resume) ?
- 6) Difference between leverage point and outlier?
- 7) Why time series different from regression?
- 8) Explained about correlated errors in time series.
- 9) Asked about matrix factorization methods such as NMF and SVD?

Additional Information

Having detailed knowledge about points mentioned in your resume gives you advantage.

Mondelez



Selected 4 Candidates

4 Selected and 1 Rejected candidates shared their experience

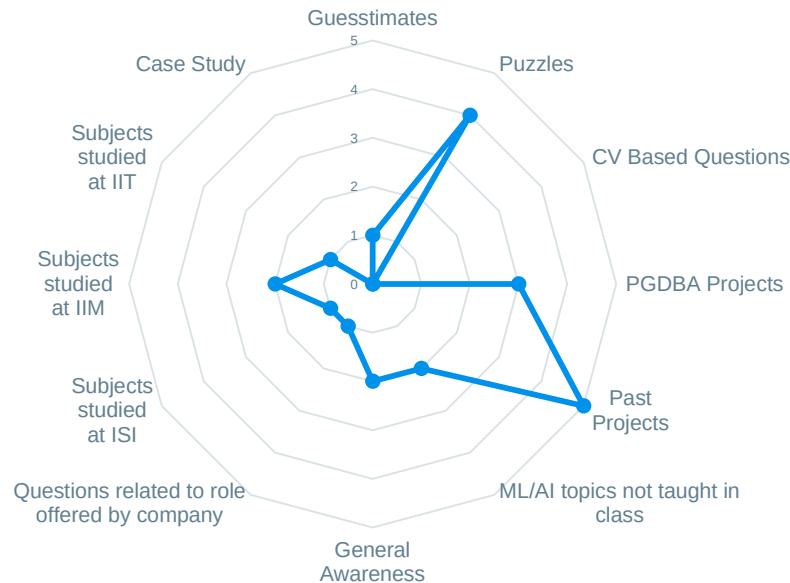




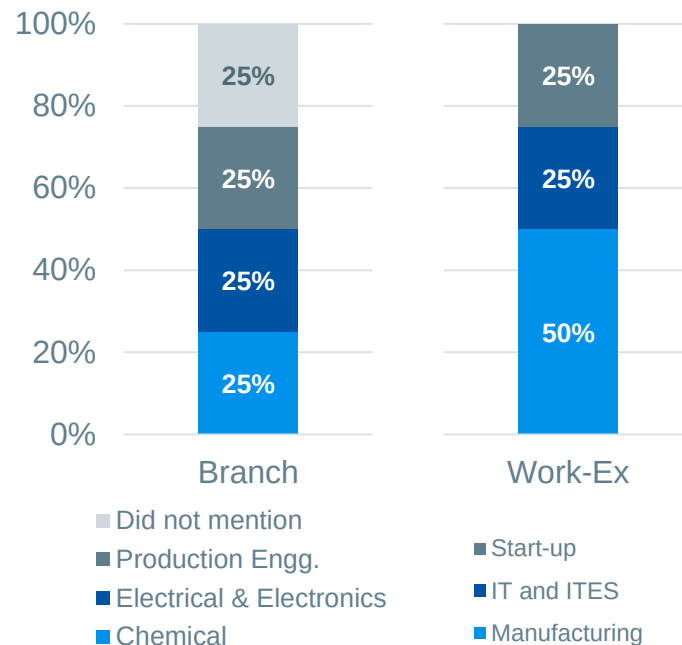
Summary

- ◎ **Shortlisting Criteria:** Group Discussion
- ◎ **Average Interview Duration:** 35 minutes

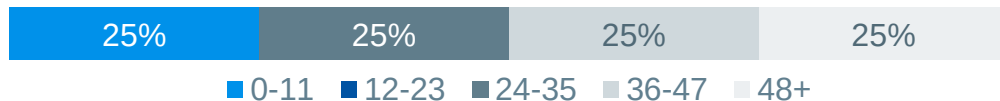
Topics asked in final interview



Profile of 4 Selected Students



Work-ex of 4 Selected Students



Sri Gowtham Nuthi

Group Discussion

Topic- Create a **Recommendation** System for sales executives on what products they have to sell, how much quantity.

Instructions- Please try to support each other's points made in the flow of GDs. Don't try to include new points, this will lead to clumsiness (which happened in almost all Mondelez GDs)

Final Interview

- 1) About work experience
- 2) About 2 projects
- 3) A bit about algorithms used in project
- 4) Guesstimate question
- 5) Wanted clarity in thought process

HR Questions

Related to Work Experience Only

Addanki Hareesh

Group Discussion

Topic - You are given suppliers transactions data and the company has outstanding balance so discuss the insights that you will get from the data and how will you improve the outstanding balance (making it low), and the data science techniques you will approach in solving this type of data

Final Interview

- 1) Regarding past work experience and about yourself they known through tell me about yourself question
- 2) Then they asked to explain any one project in detail (you need to cover what is the problem statement, how you approached ,what is the data? Any issues with the data ,different techniques you applied for the problem statement from baseline to advanced modeling and in the end the impact of the project)
- 3) Guesstimate about how to calculate how many flights fly out of Mumbai in a day?
- 4) What is your contribution in projects while you do in your course?
- 5) What is PCA?
- 6) Explain the difference between C, C++ and python?

Addanki Hareesh

Final Interview (Continued)

- 7) Computational efficiency of running a for loop in C,C++,python
- 8) What is the function of lambda function in python and why it is used?
- 9) Difference between bagging and boosting?
- 10) Splitting criteria in decision trees and random forest?
- 11) If Mondelez has certain products and if you have new products which are from Mondelez and as well as from others what features do you taken into account in image recognition what feature engineering do you do?
- 12) What type of data have you handled in work experience
- 13) How well have you used your sensory data in analysis

Addanki Hareesh

HR Questions

- 1) Why do you want to work for Mondelez?
- 2) Having worked for 4 years in manufacturing why do you shift to analytics industry?
- 3) What are the projects you are looking in Mondelez?
- 4) where do you see yourself in next 5 years?

Rest are simple questions. The HR just wants to know about me and my college, the places I have worked and asked the differences that I observed when I worked with different teams and different people

Suggestions

My observation is that you need to explain the project detail and in a very clean way

Work experience also mattered

Regarding no coding experience I covered it through achievements in hackathons

Lakshmikanta Lahamodak

Group Discussion

All were shortlisted for GD.

Topic - Building Recommendation system for Mondelez products

Total 5 questions were given to discuss: Data required, How to get data, Handling big data, What data science technique you will use, Recommendation

Final Interview

- 1) Prepare well the tell me about yourself(TMAY)
- 2) Whatever You will say/claim in TMAY, they will further ask questions from that; I have told about my proclivity towards working with data, so they asked me about my projects related to that in my past job that I have done in details. Prepare for your past projects and related key questions
- 3) One guesstimate: How many incoming and outgoing flights fly from Mumbai airports in a given day?
- 4) Asked about my two project : Home credit default risk and one Image classification project.
- 5) Mondelez has a ongoing project on **packaging logo classification**?? What data you need? How you will do this ? (*Explained transfer learning-vgg network for image classification*)

Lakshmikanta Lahamodak

HR Questions

- 1) How many interviews have you given before Mondelez? (*In a sense, why Mondelez?*)
- 2) Where do you see yourself in 5 years?

Sourav Meena *(All Interviews Compiled)*

Interview Experience:

1. Go through your Resume
2. Questions from projects and competitions
3. Image classification : CNN architecture
4. Working of LSTM and it's architecture
5. Bagging & Boosting
6. Working of tree based model Random forest and XGb
7. What is AUC, PRECISION, RECALL, F- SCORE, RMSE etc
8. Use of z-test, t-test
9. Various libraries of Python and R (only ML and visualisation)
10. Time Series - MA, ARMA, ARIMA, Seasonality, exponential smoothing, ACF and PACF plot

Karan Raj Chauhan

Group Discussion

Topic - Explain complete process to identify SKU

Final Interview

Leadership qualities

Projects: Explain project on Revenue prediction

ML: SVR vs logistic, advantage of numerical features, logistic regression

PwC DIAC

Selected 6 Candidates

4 Selected and 1 Rejected candidates shared their experience

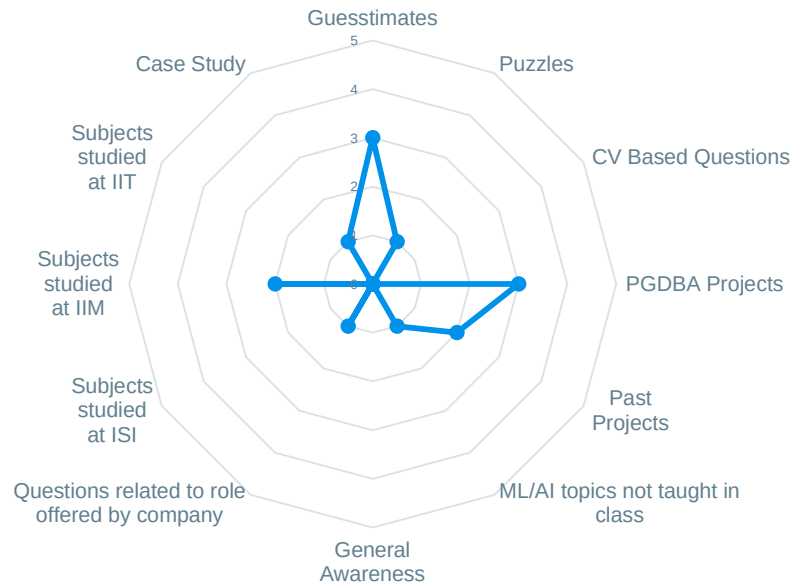




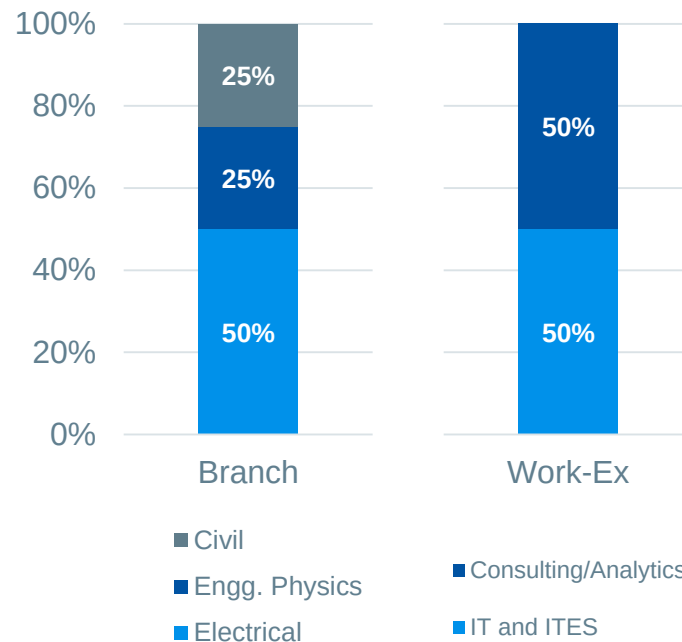
Summary

- ◎ **Shortlisting Criteria:** Resume Shortlisting
- ◎ **Average Interview Duration:** ~20 minutes
- ◎ Interview duration ranged from 5 minutes to 45 minutes.
- ◎ No specific technical rounds were there before final interview

Topics asked in final interview



Profile of 4 Selected Students



Work-ex of 4 Selected Students



Asrith Dasari

It was only a 5 minutes Interview

Final Interview

My interview lasted only for 5 to 10 minutes. There were two questions based on how would one model inflation, how do you forecast the sales in super market. One question was based on Poisson process (modelling number of accidents on Road in front of IIM Calcutta)

Suggestions

Focus on your CV

Soumik Mukherjee

Technical Round

Basics of ML, Relevant work experience, Case study

Final Interview

- 1) Walk me through your Resume
- 2) Questions related to Work-ex projects.
- 3) Difference between Random Forest Model and Decision Trees?
- 4) Case- Suppose you are told that a leading toothpaste brand faces a declining sales for the past few months (around 6 months). Asked to generate hypothesis regarding the causal factors, data sets that could be leveraged to solve the problems. Focus on business understanding of the case rather than ML tools

Suggestions

The crucial step behind PwC's process is their Resume shortlisting and Buddy program. I believe that my work-experience helped me get a shortlist and also I tried to emphasize the business impact of my projects and refrained from using excessive technical jargons/metrics in my resume for the Consult roles.

Sowrya Regana

Final Interview

- 1) Asked me explain all the projects in my CV.
- 2) Image Segmentation Project: Explain U Net Architecture, Why coding and encoding units are connected , Explain Dice coefficient to a layman
- 3) Elo Recommendation Challenge: Why Light GBM? What is Bayesian Optimization
- 4) Explain clearly about the projects (Problem Statement, Data set used, Techniques used, Evaluation Metrics)

HR Questions - Why did you leave your previous company?

Suggestions

- 1) CV making: Please don't fake the CV. Don't think using some unknown techniques/tools/anything would gain some points. Write what you know.
- 2) Learn basic coding

Arshdeep Singla

Final Interview

- 1) Logistic Regression: Is it a desirable case to have high precision & low recall OR low precision & high recall?
- 2) Work-ex related projects
- 3) Case study related questions: How will you start a Pizza Hut in IIM Calcutta campus and would it be desirable to give a 50% discount on all pizzas

Suggestions

Resume shortlisting is more or less done by all companies. Even if you don't have an analytics oriented work-ex, it would be beneficial to showcase your projects by inclining them towards analytics

Rahul Mehta

Final Interview

- 1) A retailer comes up with a 1+1 offer. Guess the change in revenue of the retailer.
- 2) What's the last movie that you watched? Guess the total revenue earned by the movie on a Friday
- 3) Elaborate on your CNN project.

Amazon BIE

Selected 3 Candidates

3 Selected and 1 Rejected candidates shared their experience

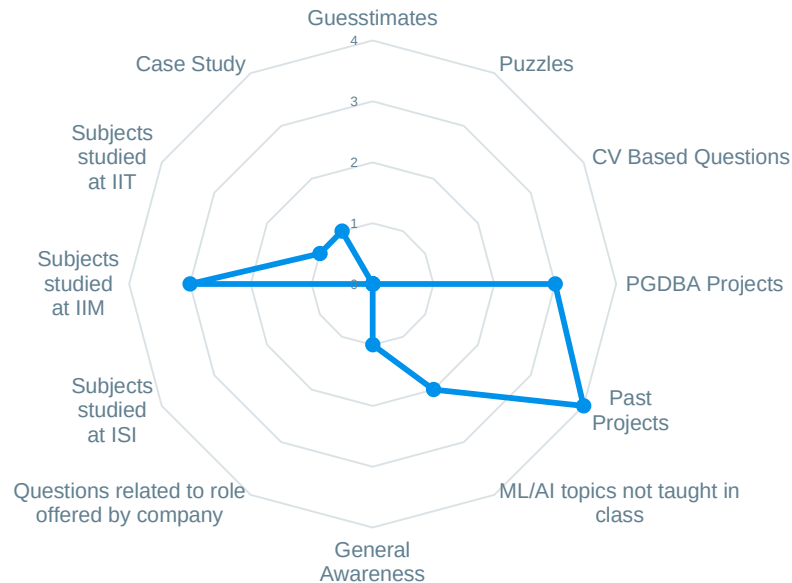




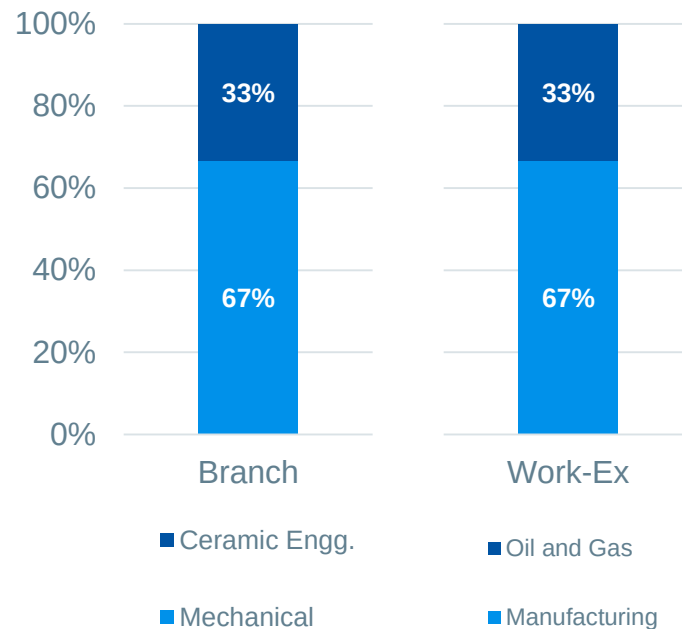
Summary

- ◎ **Shortlisting Criteria:** Resume Shortlisting
- ◎ **Average Interview Duration:** ~35 minutes
- ◎ Interview duration ranged from 20 minutes to 60 minutes.

Topics asked in final interview



Profile of 3 Selected Students



Work-ex of 3 Selected Students



Rakesh Kumar

Tech Round

- 1) Tell me about yourself?
- 2) Tell us something where you built something very simple which had huge impact and was recognized for it?
- 3) Any experience where you influenced senior management with your results
- 4) Explain any of your PGDBA Projects - I explained my Time Series Project.

Additional Comments:

Amazon focuses a lot on its 14 leadership principals and will test you on these. It is good to have basic idea about them. Don't try to fake yourself as per these principals. Be yourself and clearly justify your opinion.

Amit Bikram

Tech Round

- 1) First question was about the area of interest.
- 2) Then I was asked about a project I did on time series; Why did I chose that project, what problems I faced, what was challenging about the subject etc.
- 3) Then I was asked about CNNs and RNNs; basic questions, like convolutions, why deep learning, basic questions on RNNs, asked a couple of questions about different optimisers and their specific applications- Adam, RMSProp, Momentum
- 4) Then I was asked about stacked regression which was involved in one of my projects; explanation and application
- 5) Then I was asked to rate myself on 1-5 on python, then some questions on numpy and pandas like inbuilt functions etc. Then rate yourself on 1-5 on R: no questions on R was asked. Then I was asked to rate myself on 1-5 on SQL: then a lot of questions on primary keys, normalization, nested queries etc.

Lakshmikanta Lahamodak

Tech Round

- 1) Questions from current projects, past work projects.
- 2) Key questions like from Logistic regression -is it linear classifier? why?
- 3) why XGBoost? how do you ensure no overfitting?
- 4) SQL - how you rate yourself? are you comfortable with various joins?
- 5) The most difficult project you have worked on? where is the difficulty & how you have dealt with that? oversampling methods?? why are you ignoring the disadvantages of sampling methods?

Vyshali Nanda

Tech Round

- 1) Asked to explain any project listed in CV and then asked work ex related project.
- 2) SQL related queries.
- 3) Questions related time series , recommendation systems, metrics, clustering.

Additional Comments:

Instacart recommendation systems helped to start the discussion technically

Vikas Rai

(Information additionally given, not other stats provided)

Tech Round

- 1) Tell me about yourself?
- 2) Which project you like the most? Describe it? Follow up questions on same?
- 3) Questions on SQL. Rate yourself on python and SQL. What is difference between Distinct and Unique?

Additional Comments:

Suggestions: One page CV contents: For work experience people. After academic details Work Ex will occupy 10 lines approx, then you can accommodate maximum 6 projects(3 lines each) and 8 lines of competitions. So prepare your CV on these lines. Choose differentiating projects, use competitions as your projects. Have a look on seniors CV. Important topics Probability puzzles Python coding for shortlists Inference and FADML

Amazon DS

Selected 3 Candidates

2 Selected and 2 Rejected candidates shared their experience

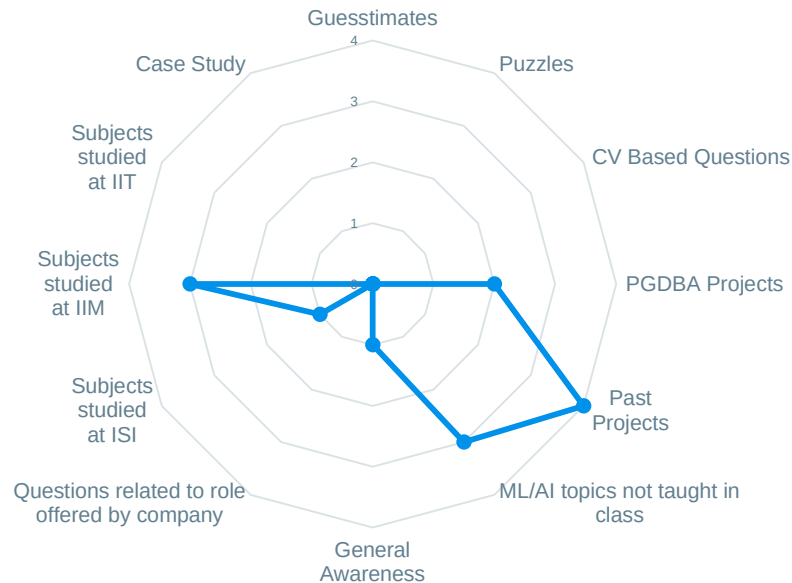




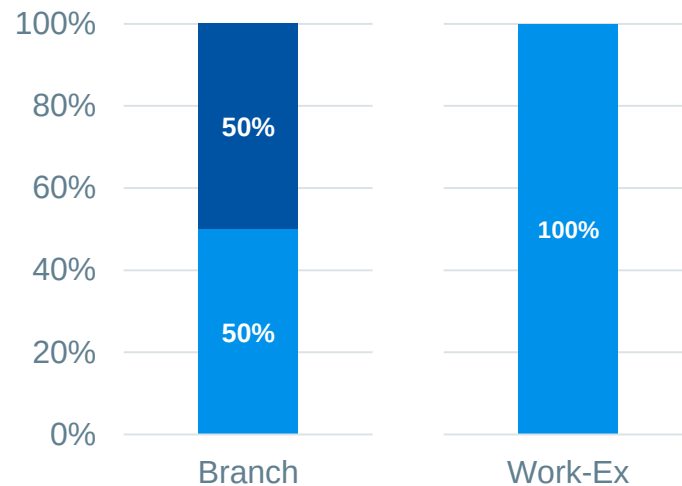
Summary

- ◎ **Shortlisting Criteria:** Resume Shortlisting
- ◎ **Average Interview Duration:** ~75 minutes
- ◎ Final Interview was majorly technical based

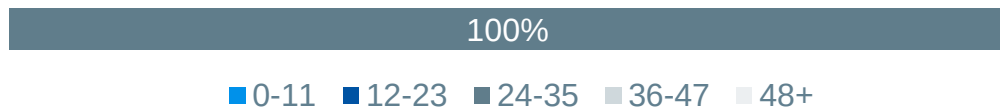
Topics asked in final interview



Profile of 2 Selected Students



Work-ex of 2 Selected Students



Akhil Kandregula

Tech Round

- 1) The interview started of with questions about the competitions listed on the resume :
 - Describe the dataset, what was the target variable, what kind of features did you create, What was the model used and why was it better vs other alternatives. Then the discussion spanned off in different directions based on the answers I was giving
 - Define logistic regression what is the model form, when is it used, why is the link function, why is a log present there?
- 2) Then was asked questions on MLE, definition and how is it implemented .Batch Gradient Descent , SGD , Minbatch GD- why one is better than the other .
- 3) Then went to Trees, Bagging vs Boosting, RF vs GBMs, GBMs vs XGBoost - What are the most important differences amongst these algos. Splitting criterion, Regularization etc .
- 4) Then he noticed the recommendation project I had showcased , started asking questions on Collaborative Filtering , Item Based Recommenders - details and implementation in matrix terminology.
- 5) A simple SQL question around joins- just the pseudo code- Employee - Supervisor Then the next 10mins went around me asking him about his experience with Amazon etc.

Akhil Kandregula

Additional Comments:

- 1) Overall , all the ML topics listed in your resume will be thoroughly tested in detail. I wasn't asked to code , but a few of my friends were asked to.
- 2) There were other ML concepts present in the resume around Graphs and Deep Learning - but weren't touched possibly in the interest of time and the comfort level of the interviewer. Hence I would recommend everyone to understand the math of all the ML algorithms and their implementation wherever possible. It is always recommended that you have diverse projects in terms of the data types (tabular, image, text, graphs etc.), data sizes analyzed, etc. to standout .

Prachi Mittal

Tech Round

- 1) Competitions Interview started with my first resume point which was a competition. I was asked questions about the data and my approach with a lot of focus on feature engineering. Various questions were asked on why I created these features and if I had some additional columns what other features would I have created and why.
- 2) Work Experience I was asked about one of the projects from my work experience in brief. Main focus was on why I chose a particular model.
- 3) Projects done during PGDBA The interviewer picked Product Recommendation from my Resume which was expected as it was Amazon. I was asked about the problem statement and approach in detail. This was followed by some general questions on recommendation like what is item-item collaborative filtering, how to approach it, how will I find similarity, how will I code it (in terms of matrices multiplication only).
- 4) Subjects at ISI I was asked about Linear Regression as I had worked on it. Basic questions on goodness of fit, outliers, leverage etc. SQL query - if I have a table of all the employees, their managers and their salaries, given an employee find his/her manager's salary.

Prachi Mittal

Tech Round

- 5) Subjects at KGP Asked me to what is logistic regression. How do we estimate coefficients, asked me derive it. Is it a closed form solution.
- 6) Tree based questions - What are decision trees, how do you decide the split, what is gini impurity, what is entropy, what is information gain, what is bagging, why do we do it, what is random forest, What is the additional step that you do in random forest, what is boosting.
- 7) ML/AI questions not taught in the course. What is gradient boosting, explain the procedure. What is XGBoost, how is it different from GBM. What is LightGBM, how is it different from XGBOOST.
- 8) Finally, asked me some basic programming questions.

Vanya Sinha

Tech Round

- 1) I was asked to explain some of my projects.
- 2) What is Bayes Theorem? Can you explain in terms of prior, posterior?
- 3) Explain any Bayesian classifier.
- 4) Puzzle: If you were stranded on an island, and knew that the probability of seeing a boat in a 30 minute window was 95%, what's the probability of seeing a boat in the next 10 min?
- 5) What's AUC? How is the ROC curve plotted?
- 6) How is a tree built? What's Gini Index?
- 7) What's the difference between bagging and boosting? Random Forest vs XGBoost vs LightGBM?
- 8) Explain MLE, MAP estimation methods.
- 9) What is the objective function of k-means? Write it down. How does it ensure "minimum intra cluster distance and maximum inter cluster distance? Is it differentiable?
- 10) Code k-means from scratch. (Actually made me code the whole thing to the T)

Shilpi Jaiswal

Tech Round

- 1) Tell me about yourself?
- 2) What was your role in the previous company?
- 3) In-depth questions on working of Logistic Regression, Adaboost, Decision Trees.
- 4) Write a code to find an element in a list, what is the complexity of the code, how can you optimize the code.

Adarsh Kumar

(Information additionally given, no other stats provided)

Tech Round

Asked 4 use cases: Two use cases in each panel.

Panel I

Case 1: Estimate the no of parcel coming to the delivery center/hub

Case 2: Estimate the time taken by the trucks from the warehouse to the city deliver center or door delivery.

Mostly related case but with certain nuances

- Create features
- Machine Learning also and why its suitable evaluation metric

Since this was the "last mile" team at amazon these questions were related to delivery.

Panel II

Case 1: Suppose a delivery is out and delivered but the customer complained that it was not delivered. How would you tackle this problem using a machine learning algorithm?

- Create product feature, user feature and user-product feature
- Suggest ML algorithm why it will work
- Select Evaluation metric

Adarsh Kumar

(Information additionally given, no other stats provided)

Case 2: Build a machine learning Algo bundle packages according to the building it has to be delivered.

Example: Suppose a delivery boy is taking package to TATA hall he should have most of the packages of TATA hall and not a few parcels belonging to TATA hall, few to MDC so on. Solve this using machine learning.

- Create features
- Suggest a machine learning algorithm

I suggested for clustering he was convinced and told me a few clustering algorithms. I think he wanted something related to Reinforcement Learning

Additional Comments:

Do all projects with utmost seriousness learning from these projects can give you cues to tackle interviews. Revise all ML algo thoroughly. Read and watch ISLR book and video multiple times to reinforce your ML understanding and ideas related to it.

VIACOM

Selected 1 Candidate

1 Selected and 2 Rejected candidates shared their experience

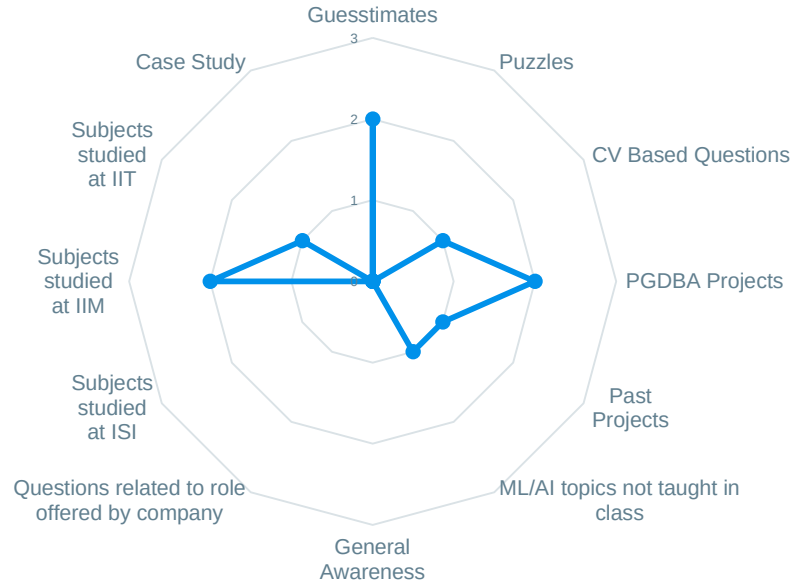




Summary

- ◎ **Average Interview Duration:** ~60 minutes
- ◎ **Interview duration of selected candidate:** 90 minutes

Topics asked in final interview



Profile of 1 Selected Students

Mechanical Engineer

Manufacturing Domain

Work-ex of 1 Selected Student

35 Months

Vikas Rai

Interview Experience:

3 panelists | 2 lady, and a senior of PGDBA

Tell me about yourself and work experience?

Case: We have library of 500 movies and we need to plan on our channel which movie to be shown at what time to generate maximum viewership? How will you do it? Write what will be hypothesis of this case , test statistic and all? And follow up questions on same.

What is Random Forest?

What is XGBoost? Difference between Random Forest and XGBoost?

What are the parameters in Logistic Regression?

One Lady panelist was intimidating, so don't lose your cool.

Vikas Rai

Suggestions:

One page CV contents:

For work experience people. After academic details Work Ex will occupy 10 lines approximately, then you can accommodate maximum 6 projects(3 lines each) and 8 lines of competitions. So prepare your CV on these lines. Choose differentiating projects, use competitions as your projects. Please go through the CVs of the seniors.

Important topics:

Probability puzzles

Python coding for shortlists

Inference and FADML

Pratik Bhanjdeo

Interview Experience:

- 1) Describe your work experience.
- 2) Why did I choose business analytics even though I have manufacturing background?
- 3) Case Study : We need to update the movie library of Viacom18. What is the approach? What all data points do I need for the problem? The problem can be divided into 2 parts: i) Build a ML model using movie performance (ratings across critics, collection), ratings generated across different time periods of the day, demographics of audience who watched in halls and in TV, etc, from the movie library already at Viacom. The y variable is ratings% (media is concerned about ratings % or TRP). Then pass your sample movie library into the model to get probable TRP that the movie can generate for different time periods. ii) On the results obtained, apply constraints such as budget to acquire rights of film, slotting of film in its predicted time slot, etc. Hence, we get probable list of movies.
- 4) Difference between RF, XGB and LGB.

Pratik Bhanjdeo

Interview Experience:

- 5) Can R^2 be negative? (It can be on test data)
- 6) Softmax function formula, Logistic regression basics, why not use linear reg for classification?
- 7) Applications of IoT,
- 8) Case Study 2 - How can we predict machine failure, what all data req. for analysis, how can we use servicing conditions to generate insights and probable actions?
- 9) A variable has only 1 value in all the rows, but is very important from business perspective? Will I keep the variable while model building? (No. Whether parametric, or non-parametric algorithm, the variable will not contribute much in model fitting)

Karan Raj Chauhan

Interview Experience:

The entire interview was based on discussions on the project (Pricing and Revenue Optimization).

Bharti Airtel

Selected 2 Candidates

2 Selected candidates shared their experience



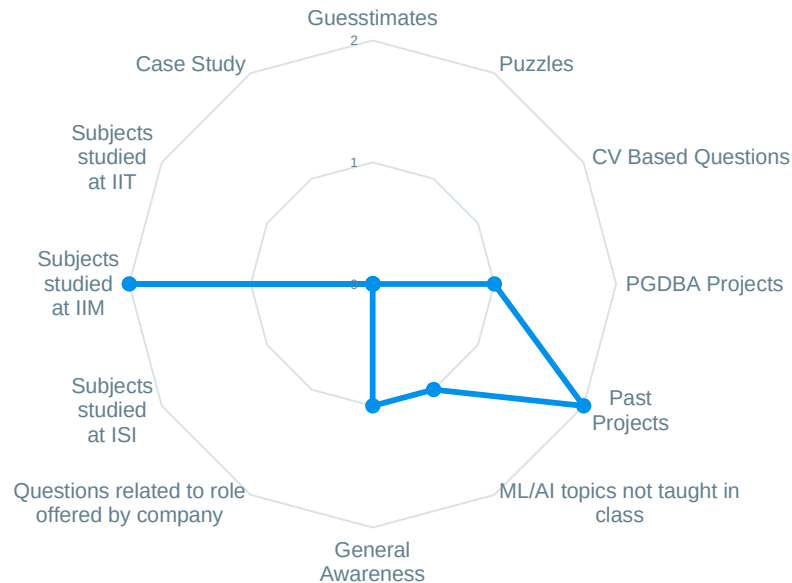


Summary

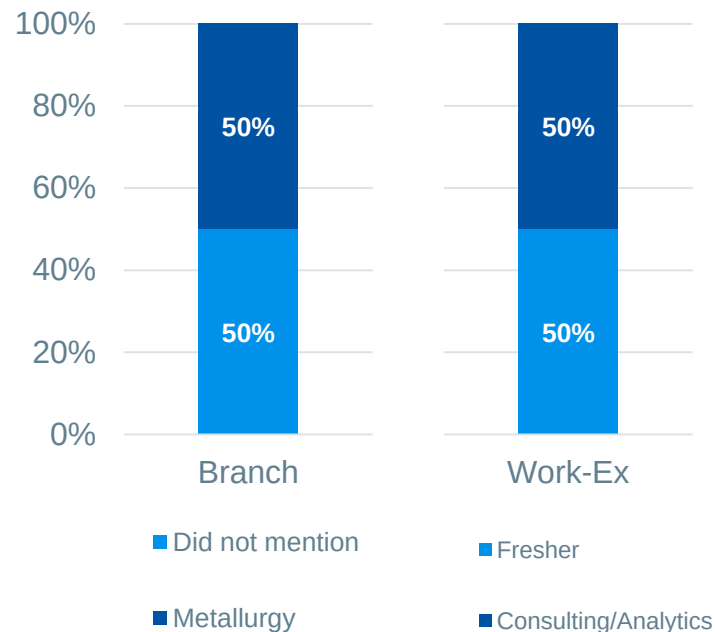
- ◎ **Shortlisting Criteria:** Online Test
- ◎ **Average Interview Duration:** ~25 minutes



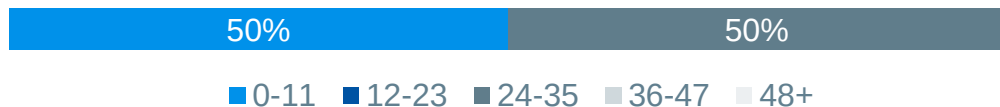
Topics asked in final interview



Profile of 2 Selected Students



Work-ex of 2 Selected Students



0-11 12-23 24-35 36-47 48+

Rajnish Kumar

Tech Round

- 1) Tell us about yourself.
- 2) Asked to explain a project which I did during PGDBA
- 3) Difference between Bagging, Boosting, Random Forest, XGBoost?
- 4) Asked about two use cases of machine learning in Telecom industry.
- 5) Asked about my project on customer churn prediction.
- 6) How you handle data imbalance? What is smote?

HR Round

- 1) Tell us about yourself.
- 2) What kind of person you are: aggressive etc.
- 3) Did you get PPO in your B.Tech internship? Why not?

Ajinkya Suryawanshi

Tech Round

Interview was mostly based on whatever you have written in your resume. They will pick any technique or project from your resume and will ask to explain about it. There will be a lot of cross questions while your explanation so you should know your resume/project in and out. Learning from ISI will definitely help. So revise your ISI semester really well.

HR Round

What kind of role/work you are looking for?

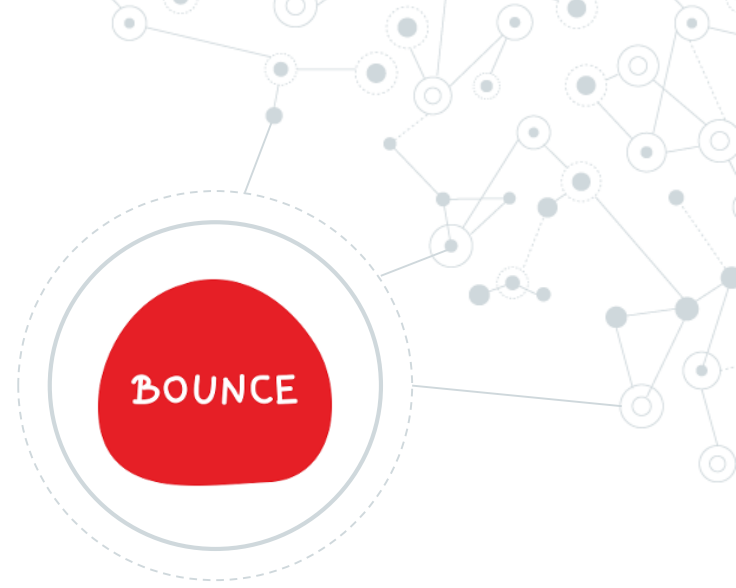
Additional Comments:

- 1) All questions asked will have hint in your resume. SO make sure you prepare your resume really well
- 2) Revise whatever being taught at ISI.

Bounce

Selected 1 Candidate

1 Selected candidate shared his/her experience

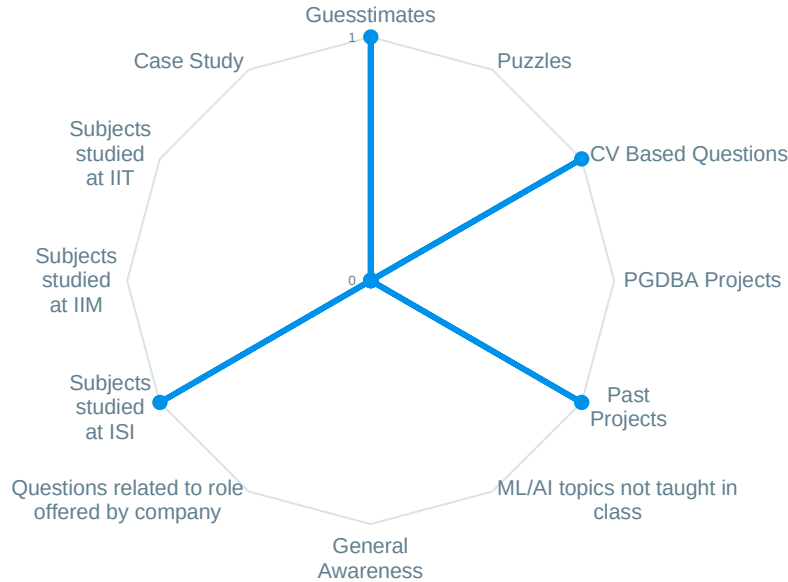




Summary

- ◎ **Entire process consisted of 5 rounds**
- ◎ **Duration of the interview: 75 minutes**

Topics asked in final interview



Profile of 1 Selected Student

Electrical Engineer

Education Domain

Work-ex of 1 Selected Student

34 Months

Vadigi Chaitanya

Technical Round - Must be well-versed with Pandas, Data Wrangling, and SQL.

Final Interview

First-Round Interview (Interviewer: AVP): It was a Skype call. Overall interview revolved around my projects I have mentioned in my resume and ability to reach the core of the problem

Second-Round Interview (Interviewer: Senior Data Scientist): It was a role-specific interview. We discussed some of the use cases at Bounce and the interviewer asked my modelling approaches, techniques to solve those problems.

Third-Round Interview (Interview panel: data scientist, HR manager): We had a conversation about one of the projects I have done, a couple of mathematical puzzles, a pseudo code for an algorithm.

Vadigi Chaitanya

HR Questions – What is the problem that we are trying to solve at Bounce?

Suggestions

Having a thorough understanding of the concepts and projects one has mentioned in his/her resume and being able to encapsulate and translate them in the short time available during the interview would improve one's chances of conversion significantly.

Johnson & Johnson



Selected 1 Candidate

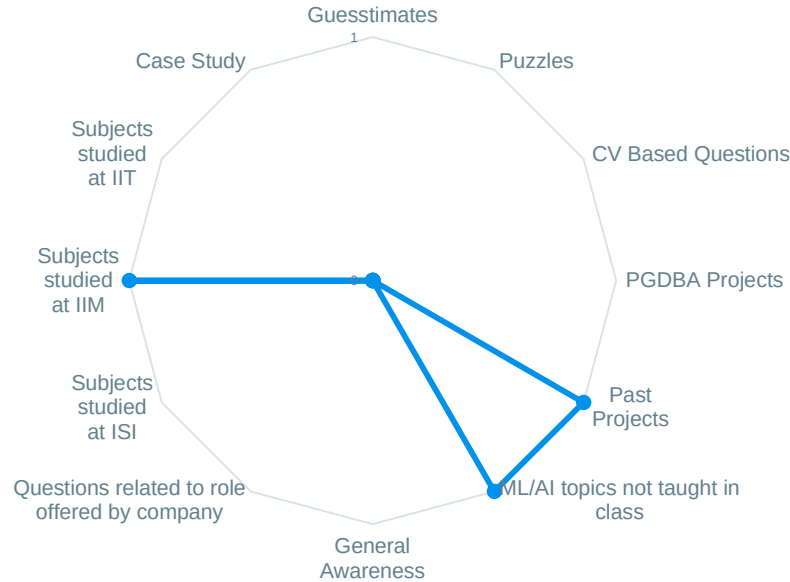
1 Selected candidate shared his experience



Summary

- ◎ **Shortlisting Criteria:** Group Discussion
- ◎ **Duration of the interview:** 15 minutes

Topics asked in final interview



Profile of 1 Selected Student

B. Tech (Branch not mentioned)
IT and ITES Domain

Work-ex of 1 Selected Student

53 Months

Adnan Tamimi

Group Discussion

GD was a mini case study. Data was provided for a company which was new entrant doing poorly in an emerging market in a particular product segment.

Tech Round 1

Questions revolved around projects done. Most of these questions came from what was written on the CV.

The questions were :

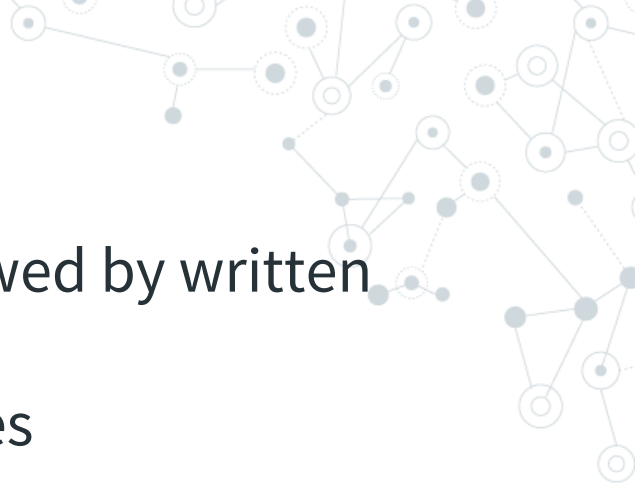
- 1) What is multicollinearity ? What to do if multicollinearity is present in the data ?
- 2) Explain what is AUC and ROC ?
- 3) How does XGBoost differ from a simple tree based model ? How is lightGBM different?
- 4) What is MAPE ?
- 5) How did stacking models help you in your project ? Why did you choose these models ?

Play Games

Selected 2 Candidates

1 Selected candidate shared his experience

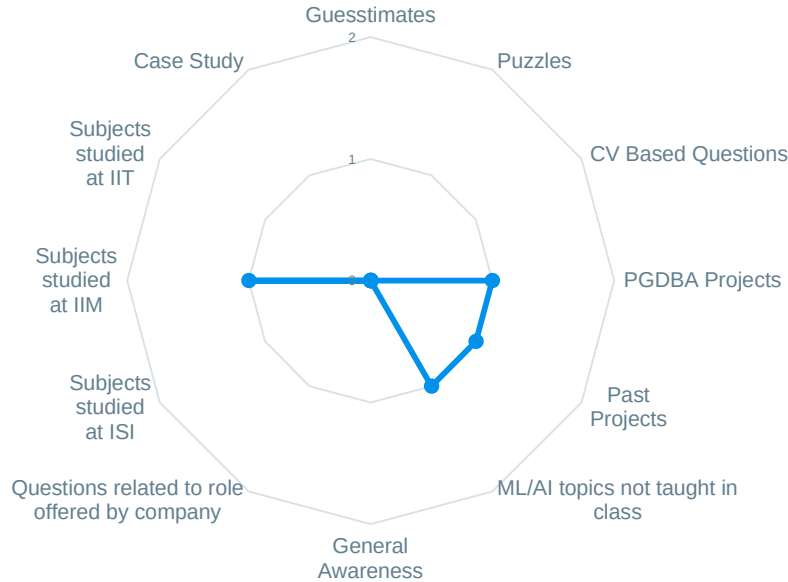




Summary

- ◎ **Shortlisting Criteria:** Interview followed by written examination
- ◎ **Duration of the interview:** 60 minutes

Topics asked in final interview



Profile of 1 Selected Student

Electrical Branch

Manufacturing Domain

Work-ex of 1 Selected Student

47 Months

Suryawansh Kumar

Written Round

There was a written test (Pen & Paper based) consisted of 2 Questions, one of them was like describe all the steps involved in any Data Science related problem where response variable is categorical & Other one was related to probability.

Technical Round

Most of the questions were related to my CV.

- 1) I had done some projects involving CNN & Transfer learning so i had been asked about these.
- 2) Difference between PCA and Factor Analysis.
- 3) Assumption of Linear Regression

Suryawansh Kumar

Final Interview Round

Interviewer - Chief Data Scientist

Nature of the question - HR & General

- 1) What's the difference between a business analyst & a data scientist ?
There isn't any perfect answer of this question but it was asked to check how you react to such shocks.
- 2) He had the answer script of the written test which they used to take some day prior to the D-day and he asked how was the test, what went well and what went wrong?
- 3) Some question related to resume like Why did you choose Data Science as specialization after Electrical Engineering?
- 4) Does this company belong to your top-4 priority in internship?
- 5) Have you published any paper/journal. Actually he was trying to test the inclination towards research side



Selected 2 Candidates

1 Selected candidate shared his experience

A network diagram consisting of a central circle with a dashed outer boundary and a solid inner boundary. Inside the circle is a dark blue rectangle containing the text "McKinsey & Company" in white. The circle is connected to a larger, complex network of nodes and lines in the top right corner of the slide. The nodes are represented by small circles, some solid and some dashed, connected by thin lines.

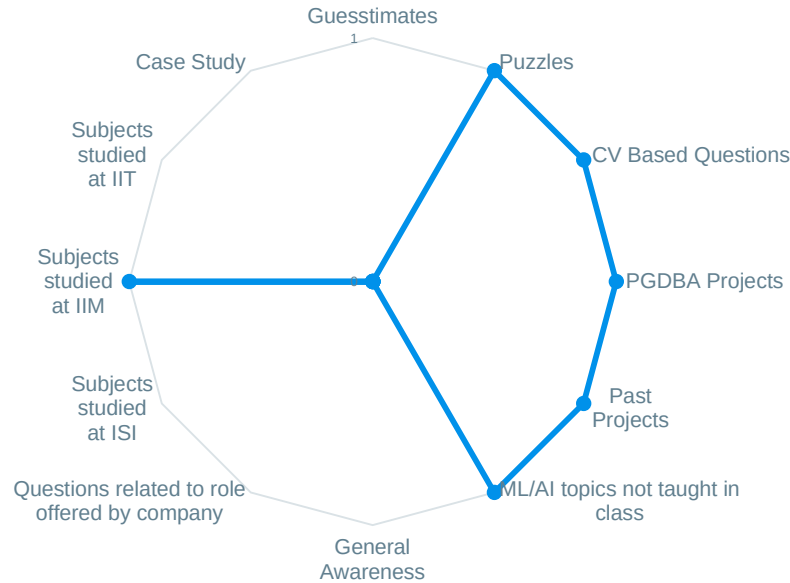
McKinsey
& Company



Summary

- ◎ **Shortlisting Criteria:** Written Test
- ◎ **Duration of the interview:** 40 minutes

Topics asked in final interview



Profile of 1 Selected Student

Electrical Branch

Teaching Domain

Work-ex of 1 Selected Student

28 Months

Sambit Behura

Tech Round

Statistics and Python

Final Interview

- 1) Two series were given and was asked to find mean and standard deviation | $\text{Var}(ax) = a^2 * \text{Var } x$
- 2) Guesstimate to find total number of cars in Kolkata .
- 3) Questions on logistic regression, complete explanation of AUC/ROC curve
- 4) How to improve AUC score by just using ensemble instead of boosting techniques (from my project)
- 5) About my experience in this course

Mentioned about the travel intensive nature of work and whether I would be comfortable doing the same

HR Questions

- 1) What was the most difficult situation you have ever been in?
- 2) How did you tackle the problems that you faced ?
- 3) Given a chance to rectify , what are the things that you could have done differently ?

The background of the slide features a light gray network pattern. It consists of numerous small circles, some of which are outlined with a dashed line, connected by thin, light gray lines. These lines form a complex web of interconnected nodes and edges, resembling a molecular structure or a data network. The pattern is dense and covers the entire background.

Thank you

This document is strictly for internal use of PGDBA Batch-5 students only. Do not share it outside the batch.